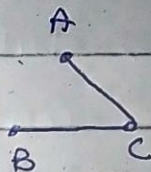
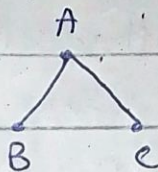
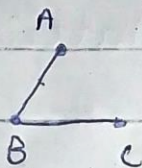
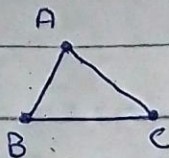


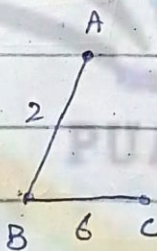
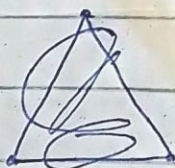
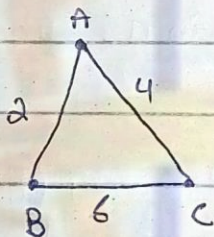
Spanning Tree

ee "An acyclic form of graph."

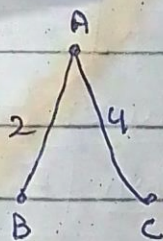


MST: Minimum Spanning Tree.

ee "Spanning Tree of a weighted graph with minimum cost edges."

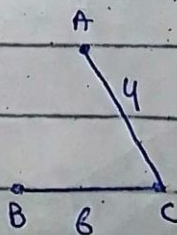


cost = 8



cost = 6

MST



cost = 10

Methods:

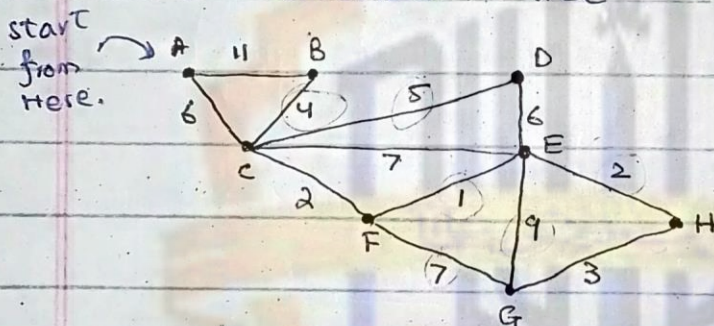
- i) Prim's MST Algorithm
- ii) Kruskal's MST Algorithm

i) Prim's MST Algorithm

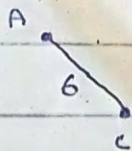
=> starts from a given node.

=> connects neighbour node with minimum cost (only if safe edge)

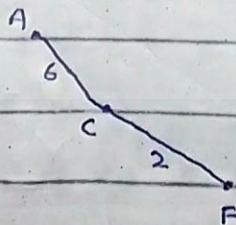
=> Connected Tree Through out process.



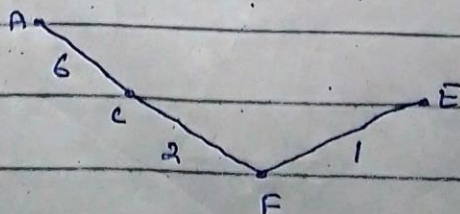
step 1: connect A to it's neighbouring node.

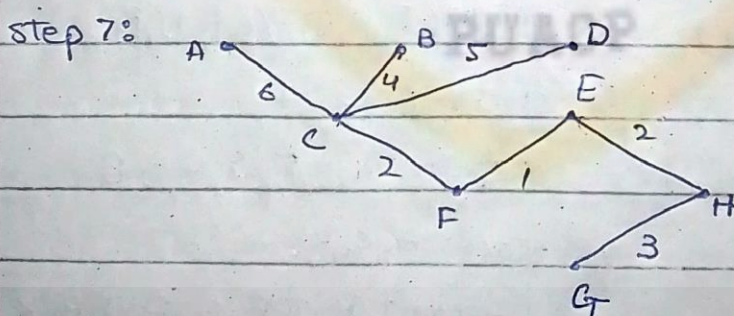
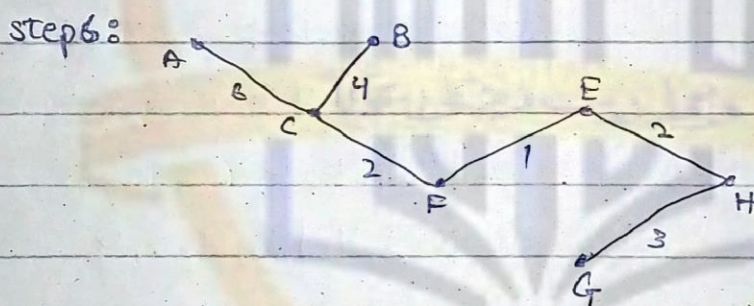
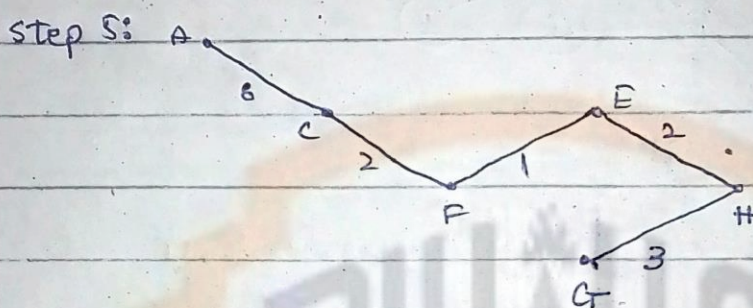
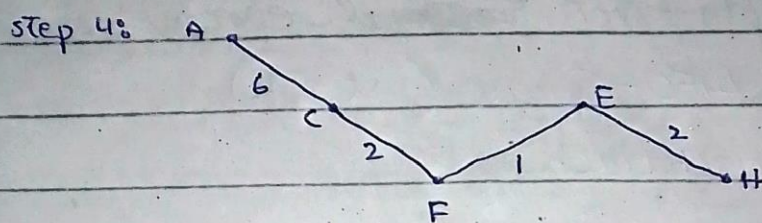


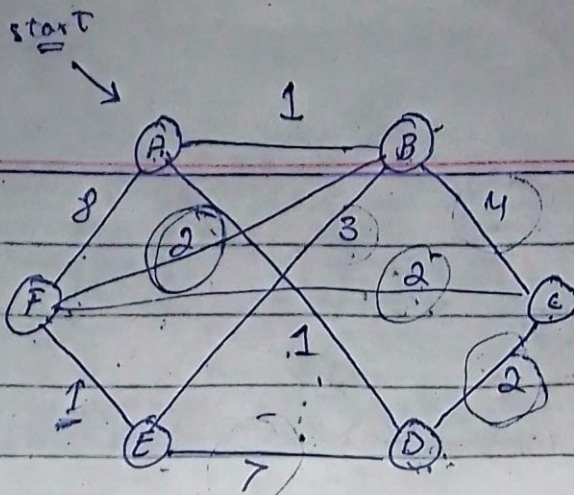
step 2:



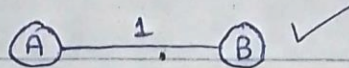
step 3:



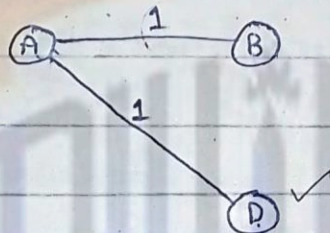




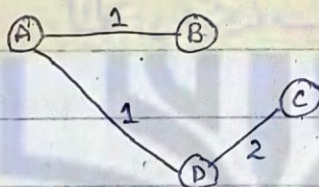
step 01:



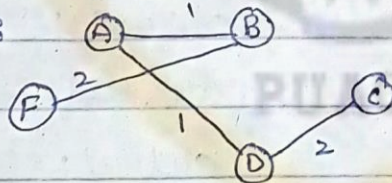
step 02:



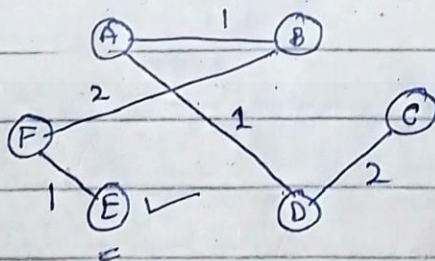
step 03:



step 04:

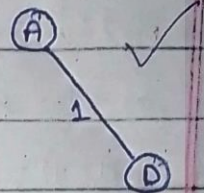


step 05:

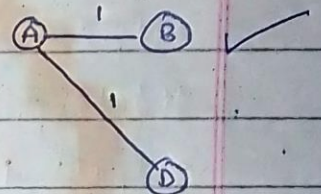


Total cost: 7

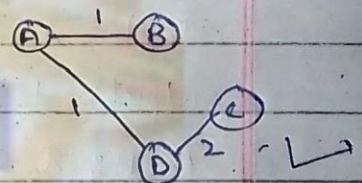
step 01:



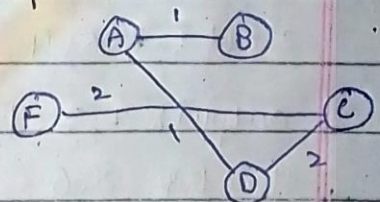
step 02:



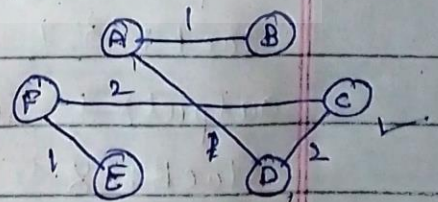
step 03:



step 04:



step 05:



Total cost: 7

With even different tree formation
Total cost remains same.